Experiment Regime Template

Twelve questions answered before the first run, and the answers for this paper

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1 What decides the design

Question 16 asks whether a 2×2-like design is necessary and what goes into that decision. One belief decides it, and in most papers that belief is already written down in the method section.

A ladder of ablations identifies the contribution of each component only when the components are additive and independent. A full factorial is the only design that estimates the interaction between them. So the question is never whether the full factorial is more thorough. It is whether you believe the two components affect each other, and you have usually said so already.

The trap is that the sentence which sounds like a reason to skip the fourth cell is the reason to run it. *The components are nested. Ablating B while keeping A does not make sense. Without B the factoring would not be clean.* Each of those says the effect of A is different depending on whether B is present, and that is what an interaction is. The belief that licenses the cheaper design is the opposite one, and it is the stronger claim: that the two components do not touch each other at all.

2 The questionnaire

Twelve questions in four blocks, answered in order, before any configuration is launched. "Not applicable" is an acceptable answer. "We will see" means the design is not chosen yet.

The claim

1. What is the claim, in a sentence a number could contradict? A claim no number can contradict does not need an experiment. It needs a rewrite.
2. Is the claim about why something works, or about which method ranks first? Why-claims survive a change of environment; a ranking on one testbed rarely does, so a design built to produce a ranking returns less than it costs.
3. Which comparison would a reader accept as contradicting the claim? Name the comparison, not the table it would sit in.

The factors

4. List every factor that could change the number, and mark each one varied, held fixed, or uncontrolled. Uncontrolled is a real category, and leaving it off the list is how a result stops replicating.
5. For each varied factor, its levels, and why those levels and not others.
6. For each fixed factor, the level it is pinned at, and the sentence the paper therefore cannot write. A held-fixed factor is a scope limit, and it reads better stated than discovered.

7. Which pairs of factors do you believe interact? Write each belief as a full sentence in the form *the effect of A is different when B is present.*

The design

8. Does any pair from question 7 contain two factors you are claiming credit for? Those pairs need every combination.
9. Does any pair contain one factor you claim credit for and one factor you are quantifying over? Then either the design crosses them, or the claim is conditional and says so.
10. What does one cell cost, in wall-clock and in whichever resource is actually scarce? For a live testbed that resource is rarely compute.

The runs and the report

11. How many seeds per cell, which statistic goes in the table, and what is released for a reader who wants a different statistic?
12. What value do you predict for each new cell, written before the run? A cell reported after the fact is a data point. A cell whose value was predicted from the mechanism is a test of the mechanism.

3 The decision rule

No pair in question 8. The ladder is sufficient, and the additivity belief goes into the paper as a stated assumption, because that is what the ladder assumes.

One pair in question 8. The full factorial over that pair. Two binary components give four cells, and three of them are usually already run.

Three or more claimed components. A resolution-IV fractional factorial recovers the main effects unconfounded with two-factor interactions at roughly half the runs.

Factors quantified over are crossed by the scope of the claim, not by this rule. If the claim says "against both attacker profiles," both appear in every reported cell.

One more line, because it is the one that gets skipped. A held-fixed factor whose level was chosen for convenience is a limitation. A held-fixed factor whose level was chosen because varying it is the follow-up project is a contribution boundary. Say which.

4 Filled, for this paper

Factor	Status	Levels	Why
Exogenous factoring	varied, claimed	on, off	first technical contribution
Opponent model	varied, claimed	on, off	second technical contribution
Learning method	varied, contrast	FOE-Dreamer, Rainbow, IQN, Expert, Random, Sleep	the sample-efficiency claim and the operational floor
Attacker profile	quantified over	Avoidant, Enticed	the claim is made against both, so both appear in every row
User process	varied in the diagnostics only	U-rand, U-burst	the two linear-decoding tables
Seed	5 per cell	5	seed-to-seed variance
Topology	fixed	8 hosts, 3 subnets	offered as a follow-up run
Defender profile	fixed	one	offered as a follow-up run
Episode structure	fixed	100 steps, 5 s polling	fixes what one interaction costs
Training budget	fixed	3 days, 1 GPU, 8 parallel environment instances	the live budget the claim is made under

Question 7, the interaction belief, is already written and already submitted. The response says the full 2×2 "was considered and set aside because the components are nested; without opponent modeling the factoring would not separate user-driven from attacker-driven variation, so the Factor+No-Opponent cell would not evaluate the factoring as designed."

That sentence answers question 8 in the affirmative. It says the factoring behaves differently depending on whether the opponent model is present, which puts the pair into the design rather than out of it. The identifiability argument in `f01` says the same thing in stronger terms: the privileged attacker supervision is the third auxiliary signal that makes the exogenous latent identifiable as user activity rather than as uncontrollable reward-relevant variation, so removing the opponent model removes the condition under which the factoring's interpretation holds. The reviewer asked for the one cell that becomes informative exactly when the nesting belief is true.

Question 12, the prediction, before the run. Read against the Avoidant column of the ablation table, additivity predicts $-78 + 16 = -62$ for Factor with no opponent. The nesting belief predicts worse than that, somewhere between -62 and -78. The sharper test is not the loss at all: fit the same ridge regression from u_t to attacker progress rather than to user activity, in both configurations, and the nesting belief predicts that it rises when the opponent model is removed. That measurement needs no new training run for the configurations already trained.

5 What the design commits you to reporting

Choosing the design settles which rows exist. It does not settle what goes in them, and the second choice is where papers in this area lose credit.

Seeds. Five per configuration is what this project has, and five is a floor rather than a target. The security systematization of deep reinforcement learning work sets a minimum of five and encourages twenty or more, and the RL methodology literature reports a case where a five-seed study of a known algorithm gave a higher mean and tighter error bars than the same algorithm at thirty seeds. Five seeds with the per-seed numbers released is defensible. Five seeds with only a mean and a standard deviation is not.

Statistic. Mean and standard deviation over seeds is what the tables carry. What makes that usable is the artifact bundle, which carries the raw per-seed episode losses so a reader can compute a bootstrap interval or an interquartile mean without asking for anything.

Convergence. The learning curves across the three-day budget are the evidence that training was not stopped early, and they belong in the body rather than an appendix. Forty-two percent of the sixty-six papers in the security systematization show none at all.

Hyperparameters. The same list every time: algorithm settings, network shape, optimizer settings, and the training procedure including the number of parallel environment instances. Thirty-two percent of that corpus reports none.

What the design is called. Five seeds on a live testbed is not a benchmark ranking and should not be written as one. It is a deployment study, which the RL methodology literature names as the setting that receives the least methodological guidance, and the setting where a paired difference against a matched baseline is worth more than a leaderboard row.

6 The cheapest sufficient version

When the design the rule selects does not fit the window, three reductions in this order, and only the third costs anything.

Measure before you train. The diagnostic that separates the two beliefs here is a regression on stored latents, not a new configuration. Any question answerable from trajectories already collected gets asked first.

Cut levels of a factor you are quantifying over before you cut a cell of a factor you claim credit for. A claim against one attacker profile is smaller and still true. A missing cell leaves the interaction unestimated, and the paper's own architecture says it is there.

Cut seeds last, and when you do, release the per-seed numbers and call the result what it is.

7 Sources

- The factorial argument, the one-factor-at-a-time defense and its four preconditions, and the resolution-IV fallback are written out in `ablation_design.tex` in this package, with its own sources: the standard design-of-experiments treatments, Frey and Wang on when one factor at a time can win, and Collins et al., *Design of Experiments with Multiple Independent Variables*.
- The identifiability argument behind the interaction belief, and the four assumptions with the two that fail, are in `f01` in this package. The predicted values for the missing cell are in `f05`.
- Patterson, Neumann, M. White and A. White, *Empirical Design in Reinforcement Learning*, JMLR 2024, for the seed-count evidence, the deployment setting, the calibration baselines, and the instruction to fix the hypothesis before the runs.

- McFadden et al., *SoK: The Pitfalls of Deep Reinforcement Learning for Cybersecurity*, for the prevalence figures and the reporting floors over its sixty-six-paper corpus.
- Agarwal, Schwarzer, Castro, Courville and Bellemare, *Deep Reinforcement Learning at the Edge of the Statistical Precipice*, NeurIPS 2021, for the aggregate statistics that a handful of seeds can still support.